

Prerequisites Assignment: Linear Algebra and Probability for Machine Learning

Part A: Linear Algebra (10 Questions)

A1. Matrix Operations: Let $A \in \mathbb{R}^{2 \times 3}$, $B \in \mathbb{R}^{3 \times 2}$. Prove or disprove: $AB \in \mathbb{R}^{2 \times 2}$ is always symmetric.

A2. Matrix Inverses: Suppose $A \in \mathbb{R}^{n \times n}$ is invertible. Prove that $(A^{-1})^T = (A^T)^{-1}$.

A3. Linear Independence: Let $v_1, v_2, v_3 \in \mathbb{R}^3$ such that $v_3 = v_1 + v_2$. Are the vectors linearly independent? Justify.

A4. Rank and Nullity: State and prove the Rank-Nullity Theorem. Give an example of a matrix where rank is less than the number of columns.

A5. Orthogonality: Let $u, v \in \mathbb{R}^n$ such that $\langle u, v \rangle = 0$. Prove that

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

A6. Eigenvalues and Symmetric Matrices: Let $A \in \mathbb{R}^{n \times n}$ be symmetric. Prove that all its eigenvalues are real.

A7. Positive Semi-definiteness: Define what it means for a matrix to be positive semi-definite (PSD). Show that for any matrix B , the matrix $B^T B$ is PSD.

A8. Change of Basis: Let $B = \{b_1, b_2\}$ be a basis for $\mathbb{R}^2$, and let $x \in \mathbb{R}^2$. Describe how to compute the coordinates of x with respect to basis B .

A9. Matrix Product AA^T : Let $A \in \mathbb{R}^{m \times n}$.

(a) Show that $AA^T \in \mathbb{R}^{m \times m}$ is symmetric.

(b) Prove that $\text{rank}(AA^T) = \text{rank}(A)$.

A10. Column Space and Row Space: Let $A \in \mathbb{R}^{m \times n}$.

(a) Define the column space and row space of A .

(b) Prove that $\dim(\text{Col}(A)) = \dim(\text{Row}(A)) = \text{rank}(A)$.

(c) Explain how the column space and row space relate to the solution set of $Ax = b$.

Part B: Probability (10 Questions)

B1. Bayes' Theorem: Derive Bayes' Theorem from the definition of conditional probability. Give a real-world example where it is applicable.

B2. Linearity of Expectation: Prove that expectation is linear:

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y],$$

even if X and Y are dependent.

B3. Variance and Independence: Let X and Y be independent random variables. Show that

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

B4. Bernoulli and Binomial Distribution: Let $X_i \sim \text{Bernoulli}(p)$, $i = 1, \dots, n$. Define $S_n = \sum_{i=1}^n X_i$.

- (a) What is the distribution of S_n ?
- (b) Derive its mean and variance.

B5. Conditional Expectation: Let $X \sim \text{Uniform}[0, 1]$ and $Y = X^2$.

- (a) Compute $\mathbb{E}[Y | X]$.
- (b) Compute $\mathbb{E}[Y]$.

B6. Law of Total Probability and Bayes: A disease affects 1% of a population. A test has 95% sensitivity and 90% specificity. Compute the probability that a randomly selected person who tests positive actually has the disease.

B7. Multivariate Gaussian Properties: Let $X \sim \mathcal{N}(\mu, \Sigma)$ where $X \in \mathbb{R}^n$.

- (a) Show that all marginal distributions of X are Gaussian.
- (b) What is the distribution of a linear projection $a^T X$? Prove it.

B8. Markov's Inequality: State and prove Markov's Inequality. Provide an example where the bound is loose.

B9. Contours of Gaussian Distributions: Let $X \sim \mathcal{N}(\mu, \Sigma)$, where $\Sigma \in \mathbb{R}^{n \times n}$ is positive definite.

- (a) Show that the contours of the pdf of X are ellipsoids centered at μ , and satisfy

$$(x - \mu)^T \Sigma^{-1} (x - \mu) = c$$

for constant c .

- (b) Explain how the eigenvalues and eigenvectors of Σ determine the shape and orientation of the ellipsoids.

(c) For $n = 2$, sketch qualitatively the contours for:

- $\Sigma = I$
- $\Sigma = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}$
- $\Sigma = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$

B10. Gaussian Conditionals: Let

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{pmatrix} \right).$$

Derive the conditional distribution $P(X \mid Y = y)$, and briefly explain its use in Bayesian inference and Gaussian Process regression.